

Cyber Kill Chain + MITRE ATT&CK Intro

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1. Objective

The goal of this task was to learn two frameworks SOC analysts use to map attacks: the Lockheed Martin Cyber Kill Chain (which breaks an intrusion into 7 sequential stages) and the MITRE ATT&CK framework (a detailed knowledge base of adversary tactics and techniques). One MITRE technique was selected and mapped onto the Kill Chain stages.

2. The 7 Cyber Kill Chain Stages

#	Stage	One-line Description
1	Reconnaissance	Attacker gathers information about the target (emails, employees, technology stack).
2	Weaponization	Attacker prepares a malware/exploit into a deliverable package, e.g. an infected document.
3	Delivery	The weaponized payload is transmitted to the target, e.g. via a phishing email, USB, or website.
4	Exploitation	The payload is triggered — a vulnerability is exploited and malicious code executes.
5	Installation	The malware installs itself on the target system, often establishing a backdoor or persistence.
6	Command & Control (C2)	The compromised system connects back to the attacker's server for remote control.
7	Actions on Objectives	The attacker accomplishes the actual goal — stealing, encrypting, or destroying data.

3. MITRE ATT&CK: Tactics vs Techniques

MITRE ATT&CK is a knowledge base documenting real-world adversary behaviour. It organises this behaviour into two key concepts:

Tactic — the “why”: the adversary's goal (e.g. Initial Access, Credential Access, Persistence).

Technique — the “how”: the specific method used to achieve that goal (e.g. Brute Force, Phishing).

Unlike the Kill Chain's fixed 7-stage timeline, MITRE ATT&CK is a much larger, more granular library covering hundreds of techniques across many tactics.

4. Selected Technique: T1110 — Brute Force

The technique T1110 (Brute Force) was researched on the official MITRE ATT&CK website.

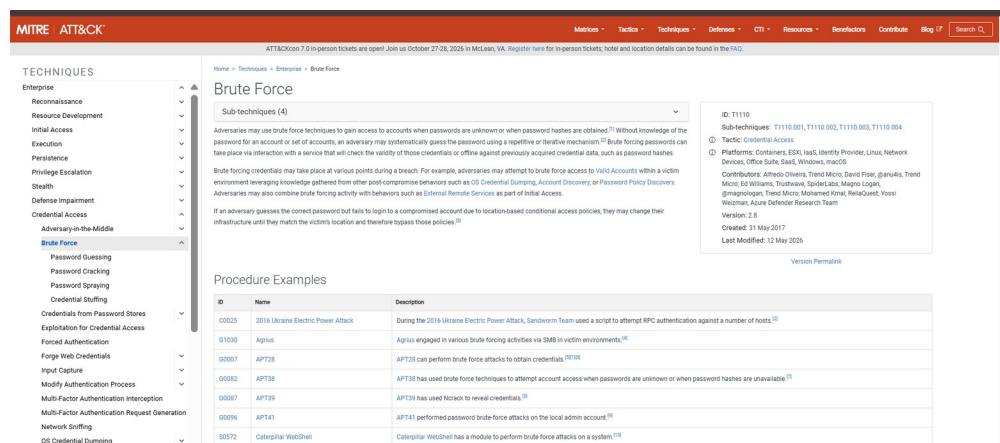


Figure 1: MITRE ATT&CK page for T1110 — Brute Force (Tactic: Credential Access).

Technique ID: T1110 · Tactic: Credential Access · Sub-techniques: Password Guessing, Password Cracking, Password Spraying, Credential Stuffing.

Description: Adversaries use brute force techniques to gain access to accounts when passwords are unknown or when password hashes have been obtained, systematically guessing credentials against a login service or offline against stolen hashes.

5. Mapping T1110 onto the Cyber Kill Chain

Kill Chain Stage	What happens under T1110 (Brute Force)
Reconnaissance	Attacker collects target usernames/emails.
Weaponization	Attacker prepares a password-guessing tool/script and a wordlist.
Delivery	Attacker begins sending login requests to the exposed service (web/RDP/SSH).
Exploitation	The correct password is found — account access is obtained.
Installation	(If applicable) a backdoor or persistence mechanism may be installed.
Command & Control	Attacker maintains ongoing access via the compromised account.
Actions on Objectives	Attacker steals data or moves laterally to other systems.

6. TryHackMe Room Completed

The required TryHackMe “Cyber Kill Chain” room was completed on the free tier, including its hands-on practical analysis of the real-world 2013 Target data breach, in which each stage of the attack (spearphishing attachment, exploiting a public-facing application, PowerShell, dynamic linker hijacking, fallback channels, and data exfiltration) was correctly mapped onto the 7 Kill Chain stages.

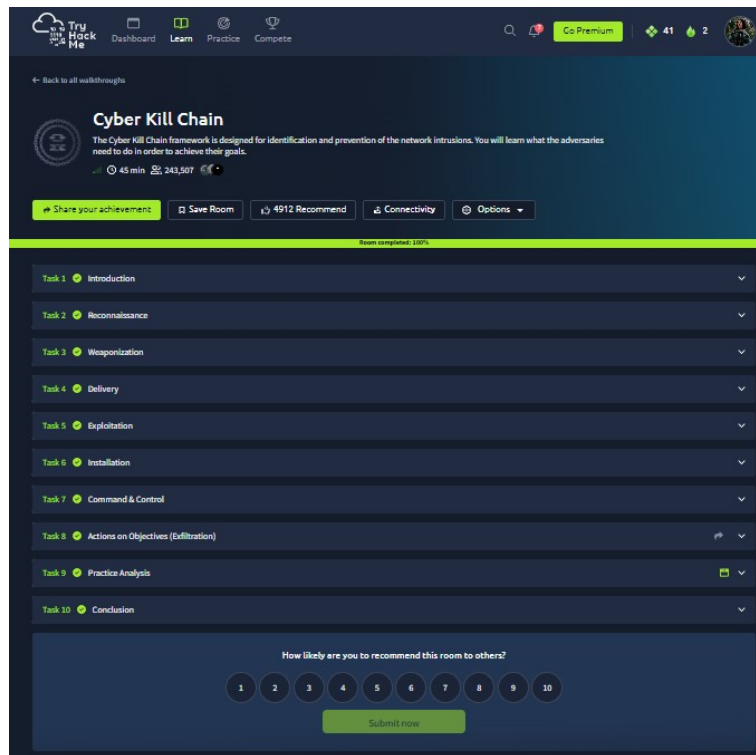


Figure 2: “Cyber Kill Chain” room — Room completed: 100%, including the Practice Analysis task.

Note: The “MITRE” room on TryHackMe is subscription-locked (Premium) and was not accessible on the free tier. As permitted by the task instructions, the official, publicly available MITRE ATT&CK website (attack.mitre.org) was used instead to research and screenshot technique T1110, in place of the locked room.

7. Conclusion

This task connected two complementary frameworks used in SOC work: the Cyber Kill Chain, which shows the sequential stages of an intrusion, and MITRE ATT&CK, which catalogues the specific tactics and techniques adversaries use at each point. Mapping a real technique (T1110 Brute Force) onto the Kill Chain, and analysing a real breach (Target, 2013) through the same lens, demonstrated how these frameworks are used together to understand and respond to attacks.